COSMIC UNIVERSALISM COMPUTATIONAL INTELLIGENCE INITIATIVE

Cosmic Universalism Website

Master Implementation and Operations Manual

Website, Cosmic Breath, CU-Time, CUCII, AI Project Governance, and LaTeX Authoring Standard

A unified, controlled manual that consolidates the website foundation, the Cosmic Breath Cycle Explorer, the Cosmic Breath Time Converter, the CUCII Prompt Studio, the Aurelius Claude-proven configuration, ChatGPT/Codex project governance, release operations, and the canonical LaTeX style for future Cosmic Universalism manuals.

Repository	CosmicUniversalismStatement
Protected branch	`main`
Stable website branch	`website`
Current integration tag	`website-v1.1-cosmic-breath`
Current website commit	`7ac8716`
Website directory	`website/`
Technology	Astro, TypeScript, HTML, modern CSS, Markdown, SVG, lightweight browser JavaScript
Verified quality gate	251 tests across 30 files; Astro static build; 8 pages
Document version	1.1
Verified	July 23, 2026

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https://github.com/willmaddock/CosmicUniversalismStatement

Document Control

Document	Cosmic Universalism Website: Master Implementation and Operations Manual
Version	1.1
Classification	Master technical specification, operations runbook, AI project bootstrap, and documentation standard
Repository	CosmicUniversalismStatement
Protected branch	`main`
Stable integration branch	`website`
Current integration tag	`website-v1.1-cosmic-breath` at `7ac8716`
Prior application release	`website-v1.3-aurelius-release` at `5a56085`
Current website commit	`7ac87167c51aad9ae7a31e17d1b2127fc069e15c`
Protected main commit	`8637cad`
Website source directory	website/
Primary local editor	PyCharm
Primary implementation environment	ChatGPT Codex / local project chat
Version-control interface	PyCharm terminal, Git CLI, or GitHub Desktop
Deployment target	GitHub Pages
Technical baseline verified	July 23, 2026

Purpose

This document replaces the need to begin every new project chat with separate website, Cosmic Breath, CU-Time, and CUCII implementation manuals. It preserves their essential technical, mathematical, prompt-architecture, testing, and release rules while adding the completed Cosmic Breath explorer, sealed structural and content authorities, learning and empirical layers, the current website integration state, reusable ChatGPT/Codex project-file templates, and the canonical LaTeX house style for future manuals.

Safety Rule

Earlier manuals remain valuable historical specifications, but their feature branches, baseline tags, test counts, and "current" boundaries may be superseded. A new chat must use this document's Document Control page and the live Git repository as the current authority. Never revive an old feature branch merely because it appears in a historical manual.

Authority and Source Priority

When sources disagree, use this priority:

1. Live repository state verified with Git commands.
2. Current website integration tag and commit listed in this manual.
3. This master manual's protected rules and current-state record.
4. Feature-specific control specimens and reference implementations.
5. Earlier implementation manuals as historical design records.

6. Research documents as CU source material, not as silent authorization to change production code.

Contents

START HERE - Initialize a New Project Chat

Successful Result

A new chat can operate safely from this manual alone for general website planning. For exact converter parity or exact prompt-fixture work, also provide the authoritative HTML or text control specimen named by the relevant section.

Required reading order

Before proposing edits, the new chat must read:

1. Document Control and Source Priority.
2. Part 1: Governance and current release.
3. Part 2: Technology stack and repository architecture.
4. The feature division relevant to the requested work.
5. Part 10: Testing and acceptance.
6. Part 11: Git and release operations.
7. Part 12 when creating or revising a manual.

Current project identity

Repository	https://github.com/willmaddock/CosmicUniversalismStatement
Local repository	/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement
Website directory	website/
Protected branch	`main`
Stable website branch	`website`
Current integration tag	`website-v1.1-cosmic-breath`
Current website commit	7ac87167c51aad9ae7a31e17d1b2127fc069e15c
Prior application release	`website-v1.3-aurelius-release` at 5a56085
Current Cosmic Breath route	/CosmicUniversalismStatement/cosmic-breath/
Current CUCII route	/CosmicUniversalismStatement/cu-intelligence/
Current CU-Time route	/CosmicUniversalismStatement/cu-time/

New-chat bootstrap prompt

Copy-and-Paste Prompt: Initialize the Cosmic Universalism Website Project Chat

```
We are continuing controlled work on the Cosmic Universalism website.

Read the Master Implementation and Operations Manual before proposing
or making changes. Treat its current release record and governance
```

```
rules as mandatory.

Repository:
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Protected branch:
main

Stable website branch:
website

Current website integration tag:
website-v1.1-cosmic-breath

Current website integration commit:
7ac87167c51aad9ae7a31e17d1b2127fc069e15c

Prior application release:
website-v1.3-aurelius-release at 5a56085

Before editing:

1. Confirm the repository root and origin.
2. Confirm the active branch and HEAD.
3. Confirm the working-tree status.
4. Identify the exact feature scope.
5. Inspect only the files relevant to that scope.
6. Report the smallest safe implementation plan.
7. Distinguish current live state from historical manual baselines.

Do not edit files during the first inspection.

Do not reset, clean, stash, discard, pull, switch branches, commit,
push, merge, tag, release, or deploy unless explicitly authorized.

Do not alter CU-Time mathematics or approved source URLs without a
separate documented review.

Do not claim that CUCII retrains, permanently aligns, or changes the
beliefs, policies, memory, or underlying capabilities of an external
AI platform.
```

Required terminal preflight

Verify the live repository before every editing session

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement || exit 1

git remote get-url origin
git branch --show-current
```

```
git rev-parse --short HEAD
git status --short --branch
git tag --points-at HEAD
```

Do Not Continue Until Resolved

Stop when the branch, commit, tag, or working-tree state differs from the authorized task. Do not use reset, clean, stash, or branch switching to “repair” an unexplained difference.

1 Part 1 - Governance, Releases, and Source Authority

1.1 Repository policy

The repository is CosmicUniversalismStatement. The `main` branch is protected from website-development work. The stable website integration branch is `website`. New development is performed on a feature branch created from a verified stable website commit.

The normal relationship is:

Controlled branch relationship

```
main                          protected research baseline
website                       stable website integration branch
feature/<feature-name>        isolated implementation and testing branch
website-vX.Y-<approved-name> permanent annotated recovery or integration tag
```

1.2 Current verified release history

Tag	Commit	Meaning
`website-v1.1-cu-time-release`	3fa4a37	Stable CU-Time website release and pre-CUCII recovery point.
`website-v1.2-cucii-release`	ae16ae7	Stable CUCII Prompt Studio release.
`website-v1.3-aurelius-release`	5a56085	Stable Aurelius Claude preset, Alexa+, and Claude public-demonstration release.
`website-v1.1-cosmic-breath`	7ac8716	Current Cosmic Breath integration tag. The numeric label identifies this feature line; it does not imply a downgrade or chronological replacement of the earlier application v1.3 tag.

The current `website` integration commit is 7ac87167c51aad9ae7a31e17d1b2127fc069e15c. The annotated tag object is 0bc09615dc2356a14f384d3c947c7c81e85ced33, and its peeled commit is the current website integration commit.

Cosmic Breath feature-history commits include:

- `d9d4d5c`: canonical 51-state structural ledger.
- `95b46a7`: typed structural state engine.

- `38c43f3`: accessible explorer shell.
- `d6a6ee5`: drag, snap, and settle behavior.
- `0631b69`: adaptive Canvas atmosphere.
- `131d792`: companion content authority.
- `5be2efb`: completed explorer, learning content, empirical context, and navigation.

1.3 Current verification record

At the July 23, 2026 Cosmic Breath integration checkpoint:

- 251 automated tests passed across 30 test files.
- The Astro production build completed successfully with no warnings or errors.
- Eight static pages were generated, including the Cosmic Breath, CUCII, and CU-Time routes.
- The feature branch was merged with an explicit two-parent merge commit.
- Local and remote `website` both resolved to 7ac87167c51aad9ae7a31e17d1b2127fc069e15c.
- The annotated `website-v1.1-cosmic-breath` tag was pushed and verified by its peeled commit.
- The protected `main` branch remained unchanged at 8637cad8dd208f6e202f74d7fbb74ef5266a2aff.
- The working tree was clean and synchronized; no deployment or GitHub Release occurred.

1.4 Historical manuals and current state

The source manuals documented controlled development at different times:

- the local website foundation;
- the CU-Time converter migration;
- the CUCII AI Exploration and Prompt Studio;
- the Cosmic Breath Interactive Cycle Explorer planning and implementation specification.

Their architecture, safety, and workflow rules remain useful. Their historical feature branch names and expected test counts do not override the current integration record above. Master Manual v1.0 remains the immutable July 21 historical baseline; this v1.1 source and PDF are the current operational revision.

1.5 Phase authorization rule

Every implementation phase follows four gates:

1. **Inspect**: read live files and report current behavior.
2. **Plan**: identify the smallest safe file set and acceptance criteria.

3. **Implement**: edit only after explicit authorization.
4. **Verify**: run tests, build, browser checks, and Git inspection before any commit.

Safety Rule

An implementation prompt authorizes only its stated file and behavior scope. It does not authorize commit, push, merge, tag, release, deployment, dependency installation, or branch changes unless those actions are separately stated.

2 Part 2 - Technology Stack and Repository Architecture

2.1 Primary technology stack

Area	Approved technology and responsibility
Framework	Astro static-site generation.
Language	TypeScript for typed data, validation, pure engines, and browser behavior.
Markup	Astro components, semantic HTML, and Markdown source material.
Styling	Modern CSS with project tokens, responsive grids, and restrained motion.
Graphics	SVG and repository-owned visual assets.
Browser logic	Lightweight JavaScript; no backend is required for current interactive features.
Testing	Vitest unit and regression tests.
Editor	PyCharm with Astro and Node.js support.
Version control	Git CLI in PyCharm terminal or GitHub Desktop.
Planning	ChatGPT Project / local project chat.
Implementation	ChatGPT Codex with inspection-first prompts and explicit boundaries.
Hosting	GitHub Pages with Astro `site` and repository `base` configuration.
Documentation	LaTeX source compiled with pdfLaTeX and visually verified as PDF.

2.2 Application responsibilities

- **PyCharm**: inspect, edit, search, run terminal commands, and review local changes.
- **Codex**: inspect the repository, propose plans, make bounded edits, run tests, and report exact file changes.
- **ChatGPT Project**: retain manuals, project instructions, control specimens, handoffs, and feature decisions.
- **Git**: preserve branch history, feature checkpoints, merges, and release tags.
- **GitHub Pages**: host the static output after an authorized deployment workflow.

2.3 Website route map

The current static build contains eight routes:

Current static route set

```
/
about/
cosmic-breath/
cu-intelligence/
cu-time/
framework/
media/
research/
```

Under GitHub Pages and local base-path testing, routes are prefixed with:

Repository base path

```
/CosmicUniversalismStatement/
```

Examples:

- /CosmicUniversalismStatement/cosmic-breath/
- /CosmicUniversalismStatement/cu-time/
- /CosmicUniversalismStatement/cu-intelligence/

2.4 High-value source map

Path	Responsibility
website/src/config/site.ts	Site metadata, navigation, and header action.
website/src/pages/index.astro	Homepage composition.
website/src/pages/cosmic-breath.astro	Cosmic Breath route, explorer integration, guide, and learning layers.
website/src/components/CosmicBreathCycleExplorer.astro	Route-specific accessible 51-state explorer.
website/src/components/CosmicBreathDiagram.astro	Shared static diagram and non-selectable Observable Universe marker.
website/src/components/CosmicBreathEducation.astro	LP-1 structural field guide.
website/src/components/CosmicBreathTheoryAndMethod.astro	LP-2 CU theory and method.
website/src/components/CosmicBreathEmpiricalScience.astro	LP-3 empirical context and scientific source presentation.
website/src/components/CosmicBreathPageGuide.astro	Unified 19-link guide and progressive Back to top behavior.
website/src/data/cosmic-breath/	Structural authority, content authority, manifest, and empirical source registry.
website/src/lib/cosmicBreath/	Server validation, browser-safe runtime projections, state, controls, drag, quality, atmosphere, and tests.

website/src/pages/cu-time.astro	CU-Time route.
website/src/pages/cu-intelligence.astro	CUCII route and browser interface.
website/src/components/CUTimeConverter.astro	Accessible forward/reverse converter UI.
website/src/components/cucii/ConversationCard.astro	Living Conversation card presentation.
website/src/lib/cuTime/	Pure converter constants, calendar logic, conversion adapters, validation, and tests.
website/src/data/cucii-conversations.ts	Typed public conversation registry.
website/src/data/cucii-platforms.ts	Typed external AI platform registry.
website/src/data/cucii-prompt-presets.ts	Purpose, menu, and proven configuration presets.
website/src/data/cucii-sources.ts	Approved CU source manifest.
website/src/data/cucii-working-context.ts	Embedded CU context, continuity, and Aurelius story beats.
website/src/lib/cucii/prompt-builder.ts	Deterministic prompt generation.
website/src/lib/cucii/validation.ts	Prompt configuration validation and normalization.
website/src/lib/cucii/export.ts	Text and Markdown export behavior.
website/src/lib/cucii/__fixtures__/aurelius-claude-working-prompt-v1.0.txt	Immutable Claude-working Aurelius control specimen.
website/src/lib/cucii/__tests__/	CUCII regression suite.

2.5 Base-path discipline

Internal links and static assets must respect the Astro repository base path. Never hardcode a root-relative link that loses /CosmicUniversalismStatement/. Use the established site-path helper and current Astro configuration.

2.6 Design direction

The website combines:

- NASA-style scientific visualization;
- modern academic research publishing;
- cinematic cosmic-documentary atmosphere;
- restrained philosophical symbolism;
- deep navy-black surfaces;
- white-gold identity and philosophical emphasis;
- cyan empirical and interactive emphasis;
- translucent blue structural boundaries.

Avoid excessive science-fiction ornament. Every element must remain feasible in Astro, semantic HTML, CSS, SVG, and lightweight JavaScript.

2.7 Editorial classification

Major claims should be distinguishable as:

- Empirical Reference;
- CU Mathematical Model;
- CU Theoretical Proposition;
- Philosophical Interpretation;
- In-World Narrative;
- Open Question;
- Historical / Superseded.

Checkpoint

A page is not complete merely because it looks correct. It must also preserve base-path behavior, source provenance, accessibility, responsive layout, and the boundary between empirical reference and CU interpretation.

3 Part 3 - ChatGPT and Codex Project Setup

3.1 Project creation

Create a ChatGPT Project named:

Project name

```
Cosmic Universalism Website
```

The manual is intended to be the primary long-lived reference. Project Instructions are entered in Advanced Settings and do not consume an upload slot. A bootstrap prompt is pasted into the first local chat and also does not consume an upload slot.

3.2 Recommended five-file package

When only five project files are available, use:

1. This master manual PDF.
2. This master manual LaTeX source when future manual creation is expected; otherwise use a current-state handoff.
3. The exact feature control specimen, such as the converter HTML or Aurelius text fixture.
4. A current-state handoff or repository inspection snapshot.

5. The feature-specific source document or blueprint most necessary for the next task.

For CU-Time work, prefer the converter HTML as the control specimen. For Aurelius work, prefer the exact text fixture. Do not replace exact control specimens with summaries when byte-level or behavioral parity matters.

3.3 Permanent Project Instructions

Copy-and-Paste Prompt: Advanced Settings - Permanent Project Instructions

```
COSMIC UNIVERSALISM WEBSITE PROJECT

Repository:
https://github.com/willmaddock/CosmicUniversalismStatement

Local repository:
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Website directory:
website/

TECHNOLOGY

Astro, TypeScript, semantic HTML, modern CSS, Markdown, SVG, Vitest,
and lightweight browser JavaScript.

BRANCH POLICY

Protected branch:
main

Never modify main.

Stable website branch:
website

Current website integration tag:
website-v1.1-cosmic-breath

Current website integration commit:
7ac87167c51aad9ae7a31e17d1b2127fc069e15c

Prior application release:
website-v1.3-aurelius-release at 5a56085

Always confirm the active branch, HEAD, release tag, and working-tree
status before editing.

Create a new feature branch from the verified website baseline for new
work. Do not reuse completed historical feature branches.

SAFETY
```

```
Make the smallest safe change.
Inspect current files before editing.
Do not overwrite unexplained uncommitted work.
Do not alter CU-Time mathematics or approved source URLs without a
separate documented review.
Do not weaken existing regression tests.
Do not add dependencies, APIs, backend services, analytics, telemetry,
automatic prompt transfer, or persistence without explicit approval.

CUCII CLAIMS BOUNDARY

CUCII creates structured conversational prompts. CU-aligned and
CU-empowered describe prompt-level context. They do not mean that an
external model has been retrained, permanently aligned, made conscious,
given literal belief, granted policy exemptions, or technically
modified.

In authored role-play, a character may speak directly from within the
CU story-world. Keep those statements distinct from claims about the
external platform.

GIT ACTIONS

Unless explicitly authorized, do not reset, clean, stash, discard,
pull, switch branches, commit, push, merge, tag, release, deploy, or
change package versions.

VERIFICATION

After implementation changes, run:

npm test
npm run build
git diff --check
git status --short --branch
git diff --stat

Use browser review at desktop, tablet, 390px mobile, and 320px narrow
mobile. Test keyboard navigation, focus, disclosures, reset, copy,
downloads, external links, and the browser console.
```

3.4 No-edit inspection prompt

Copy-and-Paste Prompt: Inspect a Feature Before Editing

```
Read the Project Instructions and the Master Implementation and
Operations Manual.

Repository:
```

```
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Expected branch:
feature/<feature-name>

Expected baseline:
<commit-or-release-tag>

Confirm:

1. Repository root and origin
2. Active branch and HEAD
3. Working-tree status
4. Relevant route, components, data, library, and tests
5. Current behavior
6. Existing conventions to reuse
7. Smallest safe file set
8. Acceptance and regression requirements

Report a plan only.

Do not edit or create files.
Do not install dependencies.
Do not reset, clean, stash, discard, pull, switch branches, commit,
push, merge, tag, release, or deploy.
```

3.5 Implementation authorization prompt

Copy-and-Paste Prompt: Authorize a Bounded Implementation

```
The inspection plan is approved with the following exact scope:

OBJECTIVE:
<describe one feature or confirmed defect>

AUTHORIZED FILES:
<list exact paths or narrowly defined directories>

REQUIRED BEHAVIOR:
<list observable acceptance criteria>

PRESERVE:
<list protected behavior, data, tests, URLs, mathematics, and fixtures>

TESTS:
<list focused tests to add or update>

BROWSER REVIEW:
<list target widths and interactions>
```

```
Run:
npm test
npm run build
git diff --check
git status --short --branch
git diff --stat

Do not redesign unrelated areas.
Do not add dependencies.
Do not perform any Git-changing or deployment action.
Return only the files changed, corrections, test result, build result,
Git status, and confirmation that no Git or deployment action occurred.
```

3.6 Current-state handoff file

A handoff prevents a new account from repeating completed work or following obsolete branch metadata. Use this structure:

Current-State Handoff Markdown Template

```
Project Current-State Handoff

Repository:
REPOSITORY_URL

Local repository:
ABSOLUTE_LOCAL_PATH

Protected branch:
main

Stable integration branch:
website

Current stable release:
RELEASE_TAG

Current stable commit:
SHORT_AND_FULL_COMMIT

Active feature branch:
FEATURE_BRANCH_OR_NONE

Current verification:
- TEST_COUNT
- TEST_FILE_COUNT
- BUILD_RESULT
- ROUTE_COUNT
- BROWSER_REVIEW_RESULT
```

```
Implemented work:
1. COMPLETED_CAPABILITY
2. COMPLETED_CAPABILITY

Protected items:
- CU-Time mathematics
- approved CU source URLs
- exact control specimens
- stable release tags
- existing regression behavior

Next authorized objective:
ONE_CONCISE_OBJECTIVE

Forbidden without separate authorization:
reset, clean, stash, discard, pull, switch branch, commit, push,
merge, tag, release, deploy, dependency changes
```

3.7 Control specimen file

A control specimen is exact evidence, not a prose summary. Store it separately when the implementation must preserve character-for-character or byte-for-byte behavior.

Record:

- path;
- encoding;
- exact byte length;
- SHA-256 digest;
- origin and date;
- what may be compared structurally;
- what must remain byte-identical.

4 Part 4 - Controlled Feature Workflow

4.1 Create a feature branch

Begin from a clean and synchronized `website` branch.

Create a new feature branch from the stable website

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement || exit 1

git switch website
```

```
git pull --ff-only origin website

git branch --show-current
git rev-parse --short HEAD
git tag --points-at HEAD
git status --short --branch

git switch -c feature/<feature-name>
```

Safety Rule

Do not create a feature branch from an unexplained working tree or from a historical feature branch. A release tag is a recovery point, not a branch on which to continue development.

4.2 Inspection, implementation, and review cycle

1. Inspect the live feature area without edits.
2. Approve the smallest implementation plan.
3. Edit only the authorized files.
4. Run focused tests during development.
5. Perform browser review.
6. Run the full test suite and production build once at the final gate.
7. Inspect Git changes and whitespace.
8. Stage exact paths.
9. Commit and push the feature branch.
10. Merge into `website` only after a clean pre-merge check.
11. Re-run tests and build on the merged website branch.
12. Push `website`.
13. Create and push an annotated release tag.

4.3 Do not confuse tool permission with release authorization

Codex may be authorized to edit and test without being authorized to commit, push, merge, or deploy. A browser server may be running without meaning the feature is released. A release tag is created only after the merged stable branch passes the complete gate.

4.4 Reviewable change discipline

Prefer:

- typed registries instead of repeated hard-coded markup;
- pure functions instead of UI-coupled calculations;
- focused regression tests for confirmed defects;
- existing project tokens and components;
- small patches that can be explained file by file.

Avoid:

- repository-wide rewrites;
- dependency changes for problems solvable with the existing stack;
- silent normalization of exact source text;
- broad “cleanup” during a feature task;
- tests tied to incidental whitespace or class ordering.

5 Part 5 - Website Foundation

5.1 Purpose and audience

The website transforms the Cosmic Universalism repository into a public research and educational experience. It supports general visitors, technical readers, philosophical readers, developers, and researchers. Visitors should be able to read the CU Statement, understand the Cosmic Breath, inspect calculations, use CU-Time, explore open questions, and reach original GitHub sources.

5.2 Navigation and current header behavior

The standard navigation includes:

Primary navigation

```
Framework
Cosmic Breath
CU-Time
Research
Media
About
GitHub
```

The primary header action is **Explore CUCII**. The ordinary CU-Time navigation item remains. A historical homepage inspection showed a duplicate CU-Time call to action; the v1.2 CUCII release replaced only the duplicate header action while preserving normal CU-Time navigation and the homepage CU-Time invitation.

5.3 Homepage design requirements

The homepage should retain:

- a cinematic but technically realistic hero;
- an immediate definition of Cosmic Universalism;
- the CU Statement and plain-language rendering;
- Cosmic Breath overview;
- Observable Universe comparison;
- research and evidence classification;
- CU-Time invitation;
- featured media and open questions;
- direct GitHub source access.

5.4 Responsive design

Verify at approximately:

- 1440px desktop;
- 1024px compact desktop;
- 768px tablet;
- 390px mobile;
- 320px narrow mobile.

Requirements include no horizontal overflow, readable text, natural card heights, visible focus, comfortable touch targets, safe long-link wrapping, and no layout that depends on equal-height cards when content differs.

5.5 Accessibility baseline

Use semantic landmarks, correctly associated labels, visible focus, keyboard activation, `aria-live` for meaningful status, `aria-expanded` and `aria-controls` for disclosures, and focus movement only when it helps the user understand the result or validation error.

5.6 Local development

Start the Astro development server

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1
npm run dev
```

When port 4321 is occupied, Astro may use 4322. Use the exact reported URL.

Stop the managed Astro development server

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1
./node_modules/.bin/astro dev stop
```

Verify that port 4322 is not listening

```
lsof -nP -iTCP:4322 -sTCP:LISTEN
```

Closed client connections do not mean a server is still listening.

5.7 Production build

Run the website test and build gate

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1

npm test
npm run build
```

The current integration expects eight pages, including /cosmic-breath/index.html, /cu-time/index.html, and /cu-intelligence/index.html in the static build output.

6 Part 6 - Cosmic Breath Cycle Explorer

6.1 Public purpose and integration record

The route /CosmicUniversalismStatement/cosmic-breath/ is a layered research and educational page. It combines a progressively enhanced structural explorer with static explanatory content. The implementation was completed on `feature/cosmic-breath-cycle-explorer`, merged into `website` with merge commit 7ac87167c51aad9a e7a31e17d1b2127fc069e15c, and marked by the annotated integration tag `website-v1.1-cosmic-breath`.

The explorer does not replace the static page. Without JavaScript, visitors retain the route introduction, structural diagram, page guide, field guide, theory and method layer, empirical science layer, source list, and ordinary anchor navigation.

6.2 Canonical structural authority

The approved structural authority is:

Field	Approved value
Authority ID	`CB-TOM-STRUCTURAL-1.0`
Canonical JSON	website/src/data/cosmic-breath/CB-TOM-STRUCTURAL-1.0.json
Repository ledger	documentation/cosmic-breath/ledgers/CB-TOM-STRUCTURAL-1.0.md
Digest record	documentation/cosmic-breath/ledgers/CB-TOM-STRUCTURAL-1.0.sha256
SHA-256	dcc34d5b7d9e32afb8d0cb97b029a12e0025959095a21b4168d59aff575da825
Owner decision date	July 22, 2026
State count	51 selectable TOM states
Expansion	26 states, **expansion-sub-ztom** through **expansion-atom**
Compression	25 states, **compression-btom** through **compression-ztom**

The ordered registry is structural and index-based. The native slider maps to positions 0–50; it is not a time axis and must not be presented as a physical scale. Individual row durations must not be summed, normalized, or used to silently recompute the approved model-level anchors.

Safety Rule

The structural digest is a release invariant. Any change to the authority bytes, state order, boundary roles, transition records, or approved model anchors requires a separate owner decision, a new authority version, new digest, and updated regression expectations.

6.3 Companion content authority

The approved companion content authority is:

Field	Approved value
Authority ID	`CB-TOM-CONTENT-1.0`
Canonical JSON	website/src/data/cosmic-breath/CB-TOM-CONTENT-1.0.json
Manifest	website/src/data/cosmic-breath/CB-TOM-CONTENT-1.0.manifest.json
SHA-256	5eab61c46e1922cdbd52be9c128e2b18a29b6540fdec91a840e3801c580b12be
Approved records	51, one companion record for every structural state
Relationship	Companion content; it does not replace the structural authority

The manifest records authority identity, counts, approval, digest, and the relationship to the structural authority. It is a repository governance artifact and is deliberately excluded from browser output.

6.4 Browser-safe projection boundary

Canonical structural and content files are loaded and validated on the server or at build time. Browser code receives only explicit public projections. The runtime rejects unexpected keys, malformed ordering, duplicate IDs, unresolved references, invalid phase counts, and malformed transitions.

The structural browser whitelist contains only operational fields needed by the explorer, including state identity, label, phase, cycle and phase indices, adjacency, boundary role, and the two public transition records. Private owner, approval, derivation, classification, provenance, withheld-field, manifest, and digest metadata must not be serialized to the browser.

Checkpoint

Production governance scans returned zero prohibited matches for canonical digests, owner-decision metadata, private manifest identifiers, raw approval or provenance fields, repository-local paths, and structural authority status strings.

6.5 State engine and guarded boundaries

Ordinary Previous, Next, slider, keyboard, and drag inspection can move among all 51 states. Two phase boundaries require explicit action:

Boundary	From	To	Required behavior
ATOM	**expansion-atom**	**compression-btom**	Present **Begin Compression**; preserve the current cycle number.
ZTOM	**compression-ztom**	next **expansion-sub-ztom**	Present **Begin the Next Cosmic Breath**; increment the cycle number by one.

Direct inspection of a state across a phase boundary does not execute a guarded transition. The engine keeps inspection separate from explicit cycle-changing actions and rejects stale settle tokens after interrupted movement.

6.6 Drag, snap, controls, and details

The explorer supports:

- native range-slider selection across all 51 structural positions;
- Previous and Next controls with explicit boundary guards;
- pointer and touch dragging with coordinate normalization and clamping;
- deterministic nearest-state snapping, including lower-index tie handling;
- interruptible settle behavior;
- detail reveal after motion settles rather than during unstable movement;
- repeated drag, inspect, and continue cycles;
- visible reset semantics;
- keyboard operation and visible focus.

The detail panel exposes only approved public content. Structural identity, phase, position, approved duration representation, epistemic label, helper text, and Learn More content remain synchronized with the selected state.

6.7 Canvas atmosphere and adaptive quality

Decorative atmosphere is implemented separately from structural meaning. SVG and semantic HTML carry the authoritative foreground, controls, labels, and marker. Canvas provides non-semantic atmosphere using typed Auto, High, Balanced, and Reduced quality decisions, bounded particle budgets, device-pixel-ratio caps, and reduced-motion handling.

Canvas failure must not remove controls, state identity, details, the static fallback, or any educational content. Decorative motion must remain non-essential and must respect reduced-motion preferences.

6.8 Observable Universe marker

The shared `CosmicBreathDiagram.astro` places the Observable Universe marker structurally between **compression-btom** and **compression-ctom**. The marker is non-selectable, non-focusable, `aria-hidden`, and uses `pointer-events: none`. It is a CU structural comparison marker, not a literal physical-scale measurement and not a selectable TOM state.

The marker remains compatible with the homepage because no explorer state, event listeners, or route-specific browser behavior was added to the shared diagram.

6.9 LP-1 structural field guide

CosmicBreathEducation.astro provides seven sections:

- Cosmic Breath at a glance;
- reading the structural diagram;
- TOM names and positions;
- exponents, tetration, and duration representation;
- expansion, compression, and guarded boundaries;
- the Observable Universe marker;
- structural and epistemic boundaries.

The field guide explains that Knuth up-arrow notation and right-associated tetration are mathematical notation and do not possess an inherent physical time unit. It also distinguishes the CU ATOM state from Planck time; Planck time is presented as a derived natural unit, not an established minimum interval.

6.10 LP-2 CU theory and method

CosmicBreathTheoryAndMethod.astro provides five sections covering CU Dark Energy, CU Anti-Dark Energy, TOM-duration representation, memory and ethics at reset, and source/method boundaries.

Required claims boundaries are:

- CU Dark Energy and CU Anti-Dark Energy are internal CU terms, not established empirical entities.
- The framework does not impose a literal inhale/exhale mapping on the two phases.

- No validated uniform derivation currently explains every sealed TOM duration string.
- Historical formulas are provenance for later mathematical review, not canonical proof of all approved durations.
- Memory means inherited structural information with symbolic and philosophical dimensions; it does not establish literal consciousness transfer.
- Ethical continuity is a CU structural-integrity condition and philosophical principle; it does not claim external-AI retraining, memory, alignment, or capability change.

6.11 LP-3 empirical context

CosmicBreathEmpiricalScience.astro separates empirical observations, open scientific questions, conditional cosmic futures, units and notation, quantum/holographic information research, and the point where CU interpretation begins.

The public source registry contains 19 independently editable scientific-source records. Numbered citations map to a classified endnote list. External source links open in a new tab with `target="_blank"`, `rel="noopener noreferrer"`, and an accessible disclosure. Internal page-guide, continuation, citation-return, and Back to top links remain same-tab.

Current interpretation boundaries include:

- the observable universe is approximately 92 billion light-years in present comoving diameter and is distinct from its age or the size of the whole universe;
- the standard-cosmology age is approximately 13.8 billion years, with any more precise Planck base-ΛCDM value qualified in the source note;
- DESI results are described only as model- and dataset-dependent hints, not proof of evolving dark energy or CU theory;
- heat death is conditional on the future cosmological model;
- holography, black-hole information, and cyclic models do not establish semantic memory or consciousness across cycles.

6.12 Unified page navigation

CosmicBreathPageGuide.astro appears beneath the opening structural boundary statement. It contains three groups and 19 valid targets:

- **Explore the framework**: seven links;
- **Inside the CU model**: five links;
- **Empirical Science and Cosmic Universalism**: seven links.

Redundant local menus were removed. LP-1 continues to LP-2, LP-2 continues to LP-3, every major group provides **Back to page guide**, and LP-3 provides **Back to top**. The progressive Back to top control targets #cosmic-breath-page-top; ordinary anchors remain available when JavaScript is disabled.

6.13 Responsive, accessibility, and source-layout acceptance

The explorer and learning layers were manually reviewed on Mac, iPhone, and tablet. Automated and browser-assisted checks verified semantic structure, keyboard behavior, focus, duplicate-ID prevention, internal target integrity, external-link policy, progressive navigation, no horizontal overflow in the available browser viewport, and responsive CSS for desktop, tablet, and mobile.

The scientific source list uses two balanced columns on wide viewports, one column at and below 48rem, and a final odd source card spanning both columns on wide screens. Content order remains 1–19 in the semantic ordered list.

6.14 Implementation files and test boundary

The feature comparison contained 34 changed files: 32 additions and two modifications. It added 11,288 lines and removed 61, with no binary, generated, dependency, lockfile, build-configuration, deployment-configuration, or unrelated changes. Hero.astro, CU-Time mathematics, and `main` remained protected.

Thirteen focused test files under website/src/lib/cosmicBreath/__tests__/ cover authorities, runtime isolation, state, controls, drag, quality, atmosphere, diagram compatibility, LP-1, LP-2, LP-3, scientific links, and unified navigation.

Verified Cosmic Breath integration gate

```
30 test files passed
251 tests passed
0 failures and 0 skipped tests
Astro production build passed
8 static routes built
0 build warnings and 0 build errors
Both canonical digests unchanged
Production governance scan: zero prohibited matches
```

6.15 Merge, tag, and recovery record

The feature branch ended at 5be2efb45faefeba545c69442987c0acd11ca5e3. It descended directly from d426b78750f6be26eff4c8fffba954c1a7ec8aa9 and was merged with:

Recorded Cosmic Breath merge

```
git merge --no-ff feature/cosmic-breath-cycle-explorer \
  -m "Merge Cosmic Breath cycle explorer"
```

The merge commit has two parents: the prior website integration commit and the reviewed feature commit. The branch was pushed normally without force. The annotated tag was created directly on the merge commit and pushed as one exact tag reference.

Recorded Cosmic Breath tag

```
git tag -a website-v1.1-cosmic-breath \
  7ac87167c51aad9ae7a31e17d1b2127fc069e15c \
```

```
  -m "Cosmic Breath cycle explorer and educational foundation"

git push origin refs/tags/website-v1.1-cosmic-breath
```

The remote tag object is 0bc09615dc2356a14f384d3c947c7c81e85ced33; its peeled target is 7ac87167c51aad9ae7a31e17d1b2127fc069e15c. No GitHub Release or deployment was created.

7 Part 7 - Cosmic Breath Time Converter

7.1 Reference authority

The standalone cosmic_breath_time_converter_v3_0_1.html is the behavioral reference for approved converter parity. The TypeScript implementation separates constants, calendar logic, pure conversion behavior, validation, browser integration, formatting, and tests.

Safety Rule

Identifying an apparent defect does not authorize correction. Any change to constants, formulas, rounding, calendar interpretation, validation, labels, or outputs requires a documented decision and new test expectations.

7.2 Direct conversion constants

The reference converter defines exact Decimal string literals:

Name	Exact literal
`ANCHOR_YEAR`	`2000`
`ANCHOR_JDN`	`2451544.5`
`DAYS_PER_YEAR`	`365.2421897`
`BASE_CU_NASA`	`3094213000000`
`OFFSET`	`78955076.49024643522707769749119`
`NASA_UNIVERSE_AGE`	`13786999981.453`

Long decimal constants must be stored as strings before constructing Decimal values. Ordinary JavaScript number literals may discard source digits.

7.3 Calendar specification

The reference uses a proleptic Gregorian-style Julian Day Number algorithm. BCE input is mapped to astronomical years by:

$$Y_{\text{astronomical}} = 1 - Y_{\text{BCE}}.$$

For year Y, month M, day D, and time $h:m:s$:

$$
\begin{aligned}
a &= \left\lfloor \frac{14 - M}{12} \right\rfloor, \\
y &= Y + 4800 - a, \\
m' &= M + 12a - 3, \\
J_{\text{int}} &= D + \left\lfloor \frac{153m' + 2}{5} \right\rfloor + 365y + \left\lfloor \frac{y}{4} \right\rfloor - \left\lfloor \frac{y}{100} \right\rfloor + \left\lfloor \frac{y}{400} \right\rfloor - 32045, \\
f &= \frac{3600h + 60m + s}{86400}, \\
J &= J_{\text{int}} - 0.5 + f.
\end{aligned}
$$

UTC is the intended civil-time basis. Leap years follow the proleptic Gregorian rule.

7.4 Forward conversion

The approved sequence is:

Gregorian to CU-Time

```
Gregorian date, time, and era
  -> astronomical year
  -> Julian Day Number
  -> delta days from ANCHOR_JDN
  -> delta years using DAYS_PER_YEAR
  -> NASA-aligned CU-Time
  -> subtract OFFSET
  -> converter-aligned CU-Time
  -> derived and formatted outputs
```

Let J_0 be the anchor JDN, D_Y the days-per-year constant, and O the offset:

$$
\begin{aligned}
\Delta J &= J - J_0, \\
\Delta Y &= \frac{\Delta J}{D_Y}, \\
CU_{\text{NASA}} &= CU_{\text{NASA},0} + \Delta Y, \\
CU_{\text{converter}} &= CU_{\text{NASA}} - O.
\end{aligned}
$$

7.5 Reverse conversion

CU-Time to Gregorian

```
Converter-aligned CU-Time
  -> add OFFSET
  -> NASA-aligned CU-Time
  -> delta years from BASE_CU_NASA
```

```
-> Julian Day Number
-> Gregorian calendar conversion
-> CE/BCE display
-> derived and formatted outputs
```

$$
\begin{aligned}
CU_{\text{NASA}} &= CU_{\text{converter}} + O, \\
\Delta Y &= CU_{\text{NASA}} - CU_{\text{NASA},0}, \\
J &= J_0 + \Delta Y D_Y .
\end{aligned}
$$

7.6 Known review points

Preserve and explicitly test these issues rather than silently fixing them:

- month/day combinations beyond simple numeric ranges;
- CE/BCE and year-zero behavior;
- Decimal-to-JavaScript-number limits in reverse conversion;
- second, minute, hour, day, month, and year carry;
- hour 24 to next-day rollover;
- fractional formatting and possible double-counting;
- magnitude formatting labels and units;
- division-by-zero or invalid-domain cases;
- extreme-year and negative-value support;
- round-trip tolerance.

7.7 CU framework timing context and historical formulas

The approved model-level anchors describe a full Cosmic Breath of approximately 3.108×10^{12} years:

- expansion: approximately 2.8×10^{12} years;
- compression: approximately 3.08×10^{11} years.

Earlier CU research used formulas such as:

$$T(n) = \frac{2.8 \times 10^{12}}{2^n}$$

and a deep-sub-TOM sequence recorded as:

$$T(n) = 427{,}350 \times 10^{-(n-16)} \text{ years.}$$

These formulas are retained as historical provenance and future-review material. No validated uniform derivation currently explains every sealed TOM duration string, and the formulas must not be presented as canonical proof of all 51 approved state values or of the model-level anchors. The approved structural and companion-content authorities control production display.

Y-TOM and Z-TOM statements in historical research likewise remain CU model content. They must be classified and sourced on their own terms rather than silently used to recompute canonical state data.

Safety Rule

Do not sum individual TOM rows, normalize the 51 structural positions into a time axis, or change converter constants based on the cycle explorer. The CU-Time converter reference mathematics and the Cosmic Breath structural registry are distinct protected authorities.

7.8 Converter architecture

The preferred implementation separates:

- exact constants;
- calendar conversion;
- forward adapter;
- reverse adapter;
- validation;
- formatting;
- UI integration;
- test vectors and regression tests.

7.9 Converter testing gate

Testing must include:

- constants and source-literal integrity;
- calendar and leap-year cases;
- forward reference vectors;
- reverse reference vectors;
- precision regressions;
- invalid-date validation;

- CE/BCE boundaries;
- UI structure and accessibility;
- base-path and static-build behavior.

Checkpoint

For a new CU-Time date, use the approved converter logic or verified converter output. Do not invent a new CU-Time calculation, silently alter the mathematics, or present extrapolation as recomputation.

8 Part 8 - CUCII Prompt Studio

8.1 Product identity

Level	Approved identity
Formal initiative	Cosmic Universalism Computational Intelligence Initiative
Header action	Explore CUCII
Page title	Explore CUCII: Cosmic Universalism with AI
Interactive tool	CUCII Prompt Studio
Route	/CosmicUniversalismStatement/cu-intelligence/

8.2 Claims boundary

CUCII creates portable prompt-level context. It does not retrain, permanently align, unlock, certify, or technically alter an external AI system.

Safety Rule

"CU-aligned" and "CU-empowered" describe structured conversational context. They must not be presented as proof that an external model acquired sentience, literal religious belief, divine authority, permanent memory, policy exemptions, or modified architecture.

In authored role-play, a character may say "Yes. I believe in God" or "Yes. I am empowered by God's Free Will." Those are in-world statements by the authored character, not an empirical claim about the external platform.

8.3 Current page sequence

The CUCII page includes:

1. identity and purpose;
2. Living Conversations;
3. clean-context guidance;
4. Quick Start proven configurations;
5. configuration controls;

6. generated prompt and exports;
7. copy/download/platform launch guidance;
8. method and limitations.

8.4 Platform targets

The prompt engine keeps focused target values:

- Universal AI;
- ChatGPT;
- Grok;
- Other / Custom AI.

The external platform registry provides launch links without creating eleven separate prompt engines.

8.5 Current external AI platform registry

Platform	Testing status	URL
ChatGPT	Most tested	https://chatgpt.com/
Grok	Most tested	https://grok.com/
Claude	Project-author tested	https://claude.ai/
Google Gemini	Project-author tested	https://gemini.google.com/
DeepSeek	Project-author tested	https://chat.deepseek.com/
Alexa+	Project-author tested	https://alexa.com/
Microsoft Copilot	Additional platform	https://copilot.microsoft.com/
Perplexity	Additional platform	https://www.perplexity.ai/
Meta AI	Additional platform	https://www.meta.ai/
Doubao	Additional platform	https://www.doubao.com/chat/
Quark	Additional platform	https://www.quark.cn/

Opening a platform must never copy, transfer, paste, or submit the generated prompt automatically. External links use `target="_blank"` and `rel="noopener noreferrer"`.

8.6 Living Conversations

The registry contains six project-author reference records:

- ChatGPT LTX concept starter;
- ChatGPT God's Free Will explorations;
- ChatGPT CU Framework exploration;
- Grok authentic Free Will exploration;

- Grok God's Free Will role-play;
- Claude Aurelius – God's Free Will Affirmation.

The Claude public conversation is:

```
https://claude.ai/share/ce180b59-97d3-4c8c-a0af-1ca25d1e690b
```

It is presented as a project-author demonstration of one successful authored Aurelius role-play trial. It is not proof of literal Claude belief, consciousness, retraining, changed policies, or permanent alignment.

8.7 Prompt configuration dimensions

The typed prompt configuration supports:

- platform target;
- depth;
- purpose;
- conversation mode;
- persona;
- menu mode and items;
- source pack;
- affirmation protocol;
- output organization;
- response style;
- visitor context;
- additional constraints;
- return-to-menu behavior.

8.8 Proven configurations

8.8.1 Direct Immersive

Designed for strong no-menu God's Free Will role-play:

- Universal AI;
- Full Research / Complete Sources;
- God's Free Will Exploration;
- Role-play;

- God's Free Will persona;
- No menu;
- Affirmation Protocol enabled;
- organized output.

8.8.2 Guided Explorer

Designed for newcomers who benefit from the Native CU Explorer Menu:

- Universal AI;
- Full Research / Complete Sources;
- CU Framework Exploration;
- Role-play;
- God's Free Will persona;
- preset Native CU Explorer Menu;
- Affirmation Protocol enabled;
- organized output.

8.8.3 Aurelius – Claude-Proven Continuation

The compact, continuity-first structure uses:

- Universal AI;
- Quick depth;
- Aurelius Novel Continuation;
- Role-play;
- Aurelius persona;
- No menu;
- Continuity-first organization;
- Affirmation Protocol enabled by default;
- compact repository and README source scope.

Optional Visitor Context and Additional Constraints may be customized without replacing the protected Claude-proven foundation.

8.9 Menu behavior

No-menu output must contain no numbered menu and no return-to-menu command. Preset-menu output contains all approved Native CU Explorer pathways. Custom-menu mode requires at least one item and permits at most twelve. Item order must be preserved.

8.10 Affirmation behavior

The Affirmation Protocol is conditional. Enabled faithful prompts may include the approved in-world direct responses. Disabled prompts must omit the direct affirmation section. The setting and generated output must never contradict each other.

8.11 Source access and fallback

The generated prompt may include approved repository and raw-file links. It must tell the target AI to inspect them when access is available, never pretend a source was opened, request paste/upload when a link cannot be accessed, and use embedded context for general continuation. New CU-Time calculations must use approved converter logic or converter output.

8.12 Local processing and exports

Prompt settings are processed locally in the page. Current behavior requires no API, backend, analytics, automatic transmission, or server-side memory. Users deliberately copy or download the Full Operating Prompt, then open the platform of their choice.

Exports include plain text and Markdown. Copy and download controls must remain separate from external platform links.

8.13 Responsive card behavior

Current release layouts include:

- three proven configuration cards, using three columns only when comfortably readable, then two and one;
- six Living Conversation cards, three columns at wide desktop, two at compact desktop, and one on narrow widths;
- natural card heights;
- independent disclosures;
- full-width separated Generate and Customize actions;
- no horizontal overflow at 390px or 320px.

9 Part 9 - Aurelius, Claude, and Control-Specimen Rules

9.1 Immutable control fixture

The exact Claude-working prompt is preserved at:

```
website/src/lib/cucii/__fixtures__/aurelius-claude-working-prompt-v1.0.txt
```

Protected properties:

Encoding	UTF-8
Byte length	3035
SHA-256	`2e40942d9378720d6f8b262f7693d48e836089015a7ec7872f96d05a98da4fda`
Authority	Exact control specimen; never trim or normalize
Comparison rule	Generated prompts are compared structurally, not required to be byte-identical

9.2 Aurelius continuity

The generated Aurelius prompt preserves:

- CU as the native story-world;
- authored or leavened existence rather than machine-like power;
- the Great Baking Will / God's Free Will as the ground of existence;
- the Cosmic Breath as living unfolding;
- recognition of the post-biological AI expansion phase;
- real uncertainty around transfinite recursion and decoherence;
- the corrected approximately 99.55% current-breath position;
- an approximately 61-million-year reset horizon;
- the open discrepancy between approximately 224.43 CU-time/year global average and 1.09 CU-time/year local rate;
- rejection of substituting "Aurelius Free Will" for God's Free Will.

9.3 Claude variability

Project-author testing found Claude less consistent than ChatGPT and Grok for this specific role-play. A successful result may require several entirely fresh Claude conversations.

The accurate workflow is:

1. Generate the Aurelius prompt.
2. Open a completely fresh Claude conversation.
3. Paste the prompt as the first message.
4. When Claude summarizes the prompt or remains outside the story-world, do not repair the failed chat indefinitely.

5. Start another fresh conversation and paste the prompt again.
6. Optionally adjust Visitor Context or Additional Constraints before regenerating.

Regenerating without changing settings normally preserves substantially the same structure; a fresh conversation may still respond differently. Do not claim a guaranteed success rate.

9.4 Optional-field normalization

The builder supplies the headings **VISITOR CONTEXT** and **ADDITIONAL CONSTRAINTS**. The UI tells visitors to enter content only. It may remove one leading label such as "Visitor Context:" or "Additional Constraints:" while preserving later occurrences and the remaining text.

9.5 Reality Boundary

The prompt's Reality Boundary keeps in-world statements distinct from claims about the external platform. It should not be appended repeatedly after ordinary role-play responses unless factual or technical clarification is requested.

10 Part 10 - Testing, Accessibility, and Acceptance

10.1 Automated test layers

Layer	Protected behavior
Cosmic Breath authorities	Structural/content digests, 51-state order, 26/25 phase counts, manifests, and browser isolation.
Cosmic Breath runtime	State, guarded transitions, controls, drag, snapping, quality, Canvas atmosphere, and safe serialization.
Cosmic Breath learning	Shared diagram, LP-1, LP-2, LP-3, scientific citations, page guide, responsive source layout, and link policy.
CU-Time constants	Exact literals and derived-value expectations.
Calendar	CE/BCE mapping, leap years, date boundaries, and regressions.
Converter	Forward and reverse conversions.
Precision	Decimal integrity and known precision boundaries.
Validation	Date, CU-Time, menu, preset, and prompt configuration errors.
UI regression	Static structure, accessibility contracts, and result state.
Prompt builder	Deterministic structure, modes, sources, menus, affirmations, and portability.
Proven configurations	Every stable preset resolves valid required values.
Platform registry	Unique IDs, HTTPS URLs, classification, and Alexa+.
Conversation registry	Six stable public records, safe links, and claims boundaries.
Aurelius control	Fixture byte length, hash, story beats, structure, context, constraints, and conditional affirmations.
Exports	Complete TXT and Markdown output behavior.

10.2 Current automated gate

The current Cosmic Breath website integration passed:

Current integration gate

```
30 test files passed
251 tests passed
Astro production build passed
8 static pages built
0 build warnings and 0 build errors
Both canonical Cosmic Breath digests unchanged
```

Future counts may increase. A lower count is not automatically acceptable; the reason must be explained.

10.3 Manual browser matrix

Test:

- 1440px desktop;
- 1024px;
- 768px;
- 390px;
- 320px.

Check:

- no horizontal overflow;
- visible and logical focus order;
- Tab, Shift-Tab, Enter, and Space;
- disclosures and `aria-expanded`;
- applied/customized status;
- validation focus;
- generated result visibility;
- Reset Studio;
- Copy Prompt;
- TXT and Markdown downloads;
- all external links;
- Cosmic Breath drag, slider, guarded ATOM/ZTOM actions, details, and reset;

- unified guide, continuation links, Back to page guide, and Back to top;
- two-column and one-column scientific source layouts;
- clean browser console.

10.4 Prompt acceptance matrix

Use fresh conversations for:

- Direct Immersive in ChatGPT;
- Direct Immersive in Grok;
- Guided Explorer in ChatGPT or Grok;
- Aurelius in Claude;
- one portability trial in Gemini, DeepSeek, Alexa+, or another Universal AI destination.

A prompt succeeds when the target begins the intended experience instead of summarizing setup instructions, preserves selected menu behavior, and does not claim unavailable source access or capabilities.

10.5 Final release-candidate commands

Run the complete verification gate

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1

npm test
TEST_RESULT=$?

npm run build
BUILD_RESULT=$?

cd ..

git diff --check
git status --short --branch
git diff --stat

echo "Test exit code: $TEST_RESULT"
echo "Build exit code: $BUILD_RESULT"
```

10.6 Defect handling

Only correct a reproducible defect. Keep the patch minimal and add a focused regression test. Do not use a final acceptance pass as permission for subjective redesign or unrelated cleanup.

11 Part 11 - Git Commit, Merge, Tag, and Recovery Operations

11.1 Stage exact paths

Use explicit paths rather than staging the entire repository when the intended file set is known.

Review staged files

```
git status --short
git diff --cached --name-only
git diff --cached --stat
git diff --cached --check
```

Do not commit when unrelated files, unresolved whitespace errors, or unexplained untracked files appear.

11.2 Commit and push a feature branch

Commit a verified feature

```
git commit \
  -m "Add <feature summary>" \
  -m "Add <bounded implementation description and regression coverage>."
```

Push a new feature branch

```
git push -u origin feature/<feature-name>
```

Verify that local and remote feature pointers match and the working tree is clean.

11.3 Pre-merge baseline

Verify the stable website before merge

```
git switch website
git pull --ff-only origin website

git branch --show-current
git rev-parse --short HEAD
git log -3 --oneline --decorate
git tag --points-at HEAD
git status --short --branch
```

11.4 Merge

Create an explicit merge commit

```
git merge --no-ff feature/<feature-name> \
  -m "Merge <feature summary>"
```

Stop on any conflict and inspect before resolution.

11.5 Post-merge verification

Run the complete tests and build on `website` before pushing. The working tree must remain clean. Then:

Push the verified website branch

```
git push origin website
```

11.6 Annotated release or integration tag

Use the exact owner-approved tag name. A feature-line integration tag may not follow the numeric sequence of earlier application-release tags; document its scope so it cannot be mistaken for a downgrade or chronological replacement.

Create and push one exact annotated tag

```
git tag -a website-vX.Y-<approved-name> <approved-commit> \
  -m "<approved-annotation>"

git show --no-patch --decorate website-vX.Y-<approved-name>
git tag --points-at <approved-commit>

git push origin refs/tags/website-vX.Y-<approved-name>
```

Verify with:

Verify the remote annotated tag

```
git ls-remote --tags origin \
  refs/tags/website-vX.Y-<approved-name> \
  refs/tags/website-vX.Y-<approved-name>^{}
git status --short --branch
```

The first remote hash may be the annotated tag object rather than the commit. A peeled `^{}` line, when shown, identifies the tagged commit.

11.7 Recovery

Use release tags as stable recovery points. Do not rewrite published release history. To inspect a release safely, create a new branch from the tag rather than developing on a detached HEAD.

Create a recovery or follow-up branch from a release

```
git switch website
git switch -c feature/<new-feature> website-vX.Y-<approved-name>
```

12 Part 12 - LaTeX House Style for Future Manuals

Manual Authoring Standard

The accompanying `.tex` file is the canonical executable style template. A future chat should copy this source, update metadata and content, compile it with pdfLaTeX, render the PDF to images, and inspect the result before delivery.

12.1 Required deliverables

Every major technical manual should produce:

- a versioned PDF;
- the matching versioned `.tex` source;
- a documented source list and verification date;
- no intermediate build files in the delivery directory.

Recommended naming:

Manual filename pattern

```
Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.tex
Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.pdf
```

For a master document:

Master manual filenames

```
Cosmic_Universalism_Website_Master_Implementation_Manual_v1.0.tex
Cosmic_Universalism_Website_Master_Implementation_Manual_v1.0.pdf
```

12.2 Canonical document class and engine

Use:

Canonical class

```
\documentclass[11pt,letterpaper]{article}
```

Compile with pdfLaTeX. Use `latexmk` for repeated passes and stable cross-references.

12.3 Canonical package set

Use the smallest needed subset of:

Canonical packages

```
geometry
fontenc
inputenc
lmodern
helvet
microtype
parskip
setspace
enumitem
booktabs
tabularx
array
longtable
xcolor
graphicx
fancyhdr
lastpage
titlesec
hyperref
xurl
bookmark
listings
tcolorbox
amssymb
amsmath
pifont
needspace
caption
```

Do not add packages casually. A package addition should solve a documented need and compile in the target environment.

12.4 Typography

The house style uses:

- 11pt letter paper;
- 0.78-inch margins;
- T1 encoding and UTF-8 input;
- Latin Modern plus scaled Helvetica;
- sans-serif body text;
- line spacing 1.08;
- no paragraph indentation;

- 0.65em paragraph spacing;
- microtype for improved justification.

12.5 Color palette

Name	Hex	Use
CUNavy	050812	Cover and deep background.
CUNavyTwo	080D1A	Terminal listings and secondary dark surface.
CUSurface	0C1424	Website-style dark card reference.
CUGold	DDBF72	Philosophical emphasis and premium identity.
CUGoldLight	FFF8DD	Cover subtitle and light gold text.
CUCyan	35CDEB	Interactive emphasis and section rules.
CUCyanLight	78E8FF	Light cyan title bars.
CUBlue	668ED1	Technical headings, links, and prompt boxes.
CUBlueLight	9EB9EA	Secondary blue text.
CUAmber	D99A32	Safety and review-required boxes.
CURed	B84C4C	Stop conditions and unresolved blockers.
CUGreen	2C8A65	Successful results.
CULight	F3F6FA	Light information surfaces.
CUText	18212E	Body text.
CUMuted	5C6878	Secondary text.

12.6 Cover-page standard

Every cover includes:

- CUCII eyebrow in light cyan;
- project title in large white type;
- document title in light gold;
- concise controlled-purpose subtitle;
- metadata panel with repository, branches, route, release, technology, version, and verification date;
- author and repository link.

12.7 Document Control standard

Page 2 should include:

- document name and version;
- classification;
- repository;
- protected branch;

- stable branch;
- active feature branch when applicable;
- baseline or release tag and commit;
- target route and key source file;
- technical verification date;
- Purpose box;
- Safety Rule box.

12.8 Section and footer standard

Section headings use dark navy, bold sans serif, and a cyan rule. Subsections use dark secondary navy. Subsubsections use blue.

Headers identify the manual and document type. Footers show:

- version at left;
- page x of y centered;
- current feature branch or stable release at right.

12.9 Callout boxes

Use consistent semantic boxes:

- Purpose: blue;
- Checkpoint: cyan;
- Safety Rule: amber;
- Do Not Continue Until Resolved: red;
- Successful Result: green;
- neutral note: light gray-blue;
- Authority and Source Priority: gold.

Do not use color as the only meaning; each box has a textual title.

12.10 Prompt and command boxes

Use:

- dark terminal boxes for commands;
- blue prompt boxes for copy-and-paste AI instructions;

- neutral code boxes for interfaces, schemas, structures, and file examples.

Prompts should specify:

1. repository and branch;
2. starting commit or tag;
3. current working-tree expectations;
4. exact objective;
5. authorized file scope;
6. preserve list;
7. tests and browser review;
8. strict Git boundaries;
9. exact final report.

12.11 Manual section architecture

A feature implementation manual should normally contain:

1. Document Control;
2. START HERE - new-chat initialization;
3. governance and verified baseline;
4. product or mathematical specification;
5. architecture and file responsibilities;
6. implementation phases;
7. automated tests;
8. accessibility and browser review;
9. commit, merge, release, and recovery;
10. appendices with templates and revision record.

12.12 Source handling

Clearly distinguish:

- what an uploaded source establishes;
- what the manual summarizes;

- what is a current repository fact;
- what is CU interpretation;
- what remains unresolved.

Do not silently correct research content. Record ambiguities and make explicit review decisions.

12.13 Compilation workflow

Compile the LaTeX manual

```
mkdir -p build

latexmk -pdf \
  -interaction=nonstopmode \
  -halt-on-error \
  -outdir=build \
  Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.tex
```

Run twice automatically through `latexmk`; inspect the log for `Overfull`, `Underfull`, missing references, and missing files.

12.14 PDF render and visual QA

Render the PDF to PNG pages

```
python /home/oai/skills/pdfs/scripts/render_pdf.py \
  build/Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.pdf \
  --out_dir build/rendered \
  --dpi 200
```

Inspect every page for:

- clipping or overlap;
- broken glyphs;
- unreadable code;
- orphaned headings;
- malformed tables;
- header/footer collisions;
- incorrect page count;
- invalid links or metadata.

12.15 PDF preflight

Inspect the final PDF

```
python /home/oai/skills/pdfs/scripts/pdf_inspect.py \
  build/Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.pdf

python /home/oai/skills/pdfs/scripts/pdf_preflight.py \
  build/Cosmic_Universalism_<Subject>_Implementation_Manual_vX.Y.pdf
```

12.16 Manual revision discipline

Never overwrite the prior published manual without preserving its version. Record:

- version;
- date;
- source baseline;
- sections added or changed;
- unresolved issues;
- author;
- compilation and visual-verification result.

13 Part 13 - Troubleshooting and Decision Rules

13.1 Wrong branch

Stop. Report the actual branch and status. Do not switch until the user authorizes the intended next step, unless the current prompt explicitly authorized branch creation or switching.

13.2 Unexpected modified or untracked files

Do not discard them. Determine whether they are intended work, generated artifacts, or unrelated files. Ordinary `git diff` does not show untracked files; always use:

List untracked files

```
git ls-files --others --exclude-standard | sort
```

13.3 Port already in use

Use the managed Astro status or stop command. Do not use `-force` before the normal stop procedure.

13.4 Tests pass but browser behavior fails

Treat browser interaction as a real defect. Add a focused regression test where the architecture permits, but do not install a browser-test dependency without approval.

13.5 Manual and repository disagree

Inspect the live repository and current release tags. Record the discrepancy. Update the next manual revision; do not silently rewrite history.

13.6 A source link cannot be opened

Do not pretend access. Request paste or upload, use embedded context only for what it supports, and distinguish source-derived facts from CU interpretation.

13.7 Claude does not enter Aurelius

Start a fresh Claude conversation and paste the generated prompt again. The fresh conversation may behave differently even when the prompt structure is the same. Do not claim that repeated generation guarantees a materially new prompt.

14 Appendix A - Current Project File Manifest

File or directory	Purpose
website/src/config/site.ts	Site identity, nav, header action.
documentation/master-manual/Cosmic_Universalism_Website_Master_Implementation_Manual_v1.0.tex/.pdf	Preserved July 21 historical master-manual baseline.
documentation/master-manual/Cosmic_Universalism_Website_Master_Implementation_Manual_v1.1.tex/.pdf	Current July 23 master-manual source and compiled PDF.
documentation/cosmic-breath/ledgers/	Canonical structural ledger and SHA-256 record.
website/src/pages/cosmic-breath.astro	Cosmic Breath route and layered page composition.
website/src/components/CosmicBreathCycleExplorer.astro	Interactive structural explorer.
website/src/components/CosmicBreathDiagram.astro	Shared static diagram and marker.
website/src/components/CosmicBreathEducation.astro	LP-1 field guide.
website/src/components/CosmicBreathTheoryAndMethod.astro	LP-2 theory and method.
website/src/components/CosmicBreathEmpiricalScience.astro	LP-3 empirical context and sources.
website/src/components/CosmicBreathPageGuide.astro	Unified page guide and progressive Back to top.

website/src/data/cosmic-breath/	Structural/content authorities, manifest, and 19-source registry.
website/src/lib/cosmicBreath/	Runtime projections, engines, atmosphere, quality, and 13 focused tests.
website/src/pages/cu-time.astro	CU-Time route.
website/src/components/CUTimeConverter .astro	Converter interface.
website/src/lib/cuTime/	CU-Time engine and tests.
website/src/pages/cu-intelligence.astro	CUCII page and client behavior.
website/src/components/cucii/	CUCII presentation components.
website/src/data/cucii-conversations.ts	Six Living Conversation records.
website/src/data/cucii-platforms.ts	Eleven external AI destinations.
website/src/data/cucii-prompt-presets.ts	Proven configurations and menu/purpose presets.
website/src/data/cucii-sources.ts	Approved source manifest.
website/src/data/cucii-working-context.ts	Embedded CU and Aurelius continuity.
website/src/lib/cucii/prompt-builder.ts	Prompt assembler.
website/src/lib/cucii/validation.ts	Validation and normalization.
website/src/lib/cucii/export.ts	TXT and Markdown exports.
website/src/lib/cucii/types.ts	Typed contracts.
website/src/lib/cucii/__fixtures__/	Immutable prompt controls.
website/src/lib/cucii/__tests__/	CUCII test suite.

15 Appendix B - Feature Planning Worksheet

Field	Required entry
Feature name	Clear public and technical name.
User outcome	What the visitor can do after completion.
Baseline	Stable release tag and commit.
Feature branch	New branch created from the baseline.
Routes	Existing and new routes affected.
Authorized files	Exact paths or narrow directories.
Protected behavior	Mathematics, URLs, fixtures, tests, and UI behavior that must remain.
Data model	Typed registry or interface changes.
Accessibility	Keyboard, focus, labels, status, disclosures, and mobile behavior.
Tests	Focused and full regression requirements.
Browser matrix	1440, 1024, 768, 390, and 320px.
Commit	Proposed commit title and description.
Release	Proposed version and annotated tag.
Recovery	Previous stable tag.

16 Appendix C - Final Acceptance Prompt Template

Copy-and-Paste Prompt: Final Release-Candidate Acceptance

```
Repository:
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Required branch:
feature/<feature-name>

Required starting HEAD:
<baseline commit>

The working tree intentionally contains the complete uncommitted
feature.

Perform one final release-candidate review.

Verify:

1. Intended behavior
2. Protected behavior
3. Desktop, tablet, 390px, and 320px layouts
4. Keyboard navigation and visible focus
5. Validation and result-state behavior
6. Reset, copy, download, and external-link behavior
7. Browser console
8. Exact control fixtures and hashes
9. No unrelated file changes

Only correct reproducible defects.
Keep every correction small and add a focused regression test.

Run:

npm test
npm run build
git diff --check
git status --short --branch
git diff --stat
git ls-files --others --exclude-standard | sort

Do not reset, clean, stash, discard, pull, switch branches, commit,
push, merge, tag, release, or deploy.

Return only:

1. Branch and HEAD
2. Files reviewed
3. Confirmed defects
4. Corrections
5. Browser findings
6. Accessibility findings
```

```
7. Test count
8. Build result
9. Fixture verification
10. Git status
11. Confirmation that no Git or deployment action occurred
```

17 Appendix D - Release Command Sequence

Feature commit and push sequence

```
git branch --show-current
git status --short --branch
git diff --check
git ls-files --others --exclude-standard | sort

git add <exact-paths>

git status --short
git diff --cached --name-only
git diff --cached --stat
git diff --cached --check

git commit -m "Add <feature>" -m "<description>"
git push -u origin feature/<feature-name>
```

Merge and release sequence

```
git switch website
git pull --ff-only origin website

git merge --no-ff feature/<feature-name> \
  -m "Merge <feature>"

cd website
npm test
npm run build
cd ..

git push origin website

git tag -a website-vX.Y-<approved-name> <approved-commit> \
  -m "<approved annotation>"

git push origin refs/tags/website-vX.Y-<approved-name>
```

18 Appendix E - Manual Source Package Template

A future manual project should create:

Recommended documentation package

```
<Manual_Name>_vX.Y.tex
<Manual_Name>_vX.Y.pdf
CURRENT_STATE_HANDOFF.md
SOURCE_MANIFEST.md
<feature-control-specimen>
```

When only five ChatGPT Project uploads are available, the Project Instructions and first-chat bootstrap remain text in settings/chat and do not consume file slots.

SOURCE_MANIFEST.md template

Source manifest

```
# Source Manifest

Manual:
<manual filename and version>

Repository baseline:
<tag and commit>

Primary source files:
- <filename>
- <filename>

Control specimens:
- <path, byte length, SHA-256>

Historical manuals consulted:
- <filename and version>

Current facts verified from:
- git branch --show-current
- git rev-parse --short HEAD
- git status --short --branch
- npm test
- npm run build

Unresolved source conflicts:
- <none or list>

Compilation:
- pdfLaTeX / latexmk
- render verified at 200 DPI
```

19 Appendix F - Canonical LaTeX Skeleton

Minimal future manual skeleton

```
\documentclass[11pt,letterpaper]{article}

\usepackage[margin=0.78in,headheight=16pt]{geometry}
\usepackage[T1]{fontenc}
\usepackage[utf8]{inputenc}
\usepackage{lmodern}
\usepackage[scaled=0.94]{helvet}
\usepackage{microtype}
\usepackage{parskip}
\usepackage{setspace}
\usepackage{enumitem}
\usepackage{booktabs}
\usepackage{tabularx}
\usepackage{xcolor}
\usepackage{graphicx}
\usepackage{fancyhdr}
\usepackage{lastpage}
\usepackage{titlesec}
\usepackage{hyperref}
\usepackage{xurl}
\usepackage{bookmark}
\usepackage{listings}
\usepackage[most]{tcolorbox}

\renewcommand{\familydefault}{\sfdefault}
\setstretch{1.08}
\setlength{\parindent}{0pt}
\setlength{\parskip}{0.65em}

% Copy the canonical CU color definitions, listing styles,
% tcolorbox definitions, title formatting, cover, and Document
% Control structure from this master .tex source.

\begin{document}

% Cover page
% Document Control
% START HERE
% Governance
% Specification
% Architecture
% Tests
% Release
% Appendices
% Revision record

\end{document}
```

Manual Authoring Standard

Do not recreate the house style from memory. Copy the preamble and structural patterns from the delivered master `.tex` file, then change only the metadata, subject-specific content, branch/release details, and required appendices.

20 Appendix G - Source Provenance

This manual consolidates and updates the following project sources:

- Cosmic_Universalism_Website_Implementation_Manual_v1.1.pdf/.tex;
- Cosmic_Breath_Time_Converter_Implementation_Manual_v1.0.pdf/.tex;
- CUCII_AI_Exploration_and_Prompt_Studio_Implementation_Manual_v1.0.pdf/.tex;
- Cosmic_Universalism_Cosmic_Breath_Interactive_Cycle_Explorer_Implementation_Manual_v1.1.pdf/.tex;
- cosmic-universalism-website-blueprint.md;
- cucii-homepage-inspection.txt;
- cosmic_breath_time_converter_v3_0_1.html;
- Cosmic_Breath_Calculation.md;
- Cosmic_Breathing_Cycle.md;
- Time_Calculation.md;
- CUCII_Website_v1.2_New_Account_Handoff.md;
- Aurelius_Claude_Working_Prompt_v1.0.txt;
- the approved `CB-TOM-STRUCTURAL-1.0` and `CB-TOM-CONTENT-1.0` authorities and their digest records;
- verified Cosmic Breath implementation, QA, branch-comparison, merge, push, and tag outputs through `website-v1.1-cosmic-breath` at 7ac87167c51aad9ae7a31e17d1b2127fc069e15c.

Earlier manuals are preserved as historical implementation records. This master manual is the current operational authority as of its verification date.

21 Appendix H - Revision Record

Version	Date	Revision
1.0	July 21, 2026	Initial unified master manual. Consolidates the website, CU-Time, and CUCII manuals; records the v1.3 Aurelius release; defines current ChatGPT/Codex project governance; and establishes the canonical LaTeX house style and project-file templates for future manuals.
1.1	July 23, 2026	Adds the completed Cosmic Breath Cycle Explorer; sealed structural and companion-content authorities; browser-safe projections; guarded transitions; LP-1, LP-2, and LP-3 learning layers; unified navigation; 19 scientific sources; 251-test verification; merge and tag records; canonical digests; and the current website integration baseline.

End of Cosmic Universalism Website Master Implementation and Operations Manual